Computer and 2-D Robot Arm Control by sEMG signals

Group 1 : Mohamed Tawakol Ahmed , Paolo Nicolini , Endri Gjinaj

Results

Random Task results

Command window output

```
Loaded Cometa, random task: 241592 samples = 120.8 s, 21 targets
Muscles: R biceps, L biceps, L trapezius, R trapezius

Part 1 - Correlation and lag values
Correlation coefficient for R biceps [0 - 1] = 0.96
MCI lags our R biceps signal by 38.50 milliseconds
Correlation coefficient for L biceps [0 - 1] = 0.94
MCI lags our L biceps signal by 37.50 milliseconds

Correlation coefficient for L trapezius [0 - 1] = 0.83
MCI lags our L trapezius signal by 39.50 milliseconds

Correlation coefficient for R trapezius [0 - 1] = 0.88
MCI lags our R trapezius signal by 39.00 milliseconds

Part 2 - MCI map
x speed = 50 * (R biceps - L biceps)
y speed (down) = 50 * (R trapezius - L trapezius)

Part 3 - Cursor mean Errors
Our error mean = 0.85 cm
MCI error mean = 0.43 cm

Part 5 - Robot arm tracking
Mean tracking error = 0.08 mm
Max tracking error = 30.37 mm
Min |det(J)| = 0.0591
```

Figures:

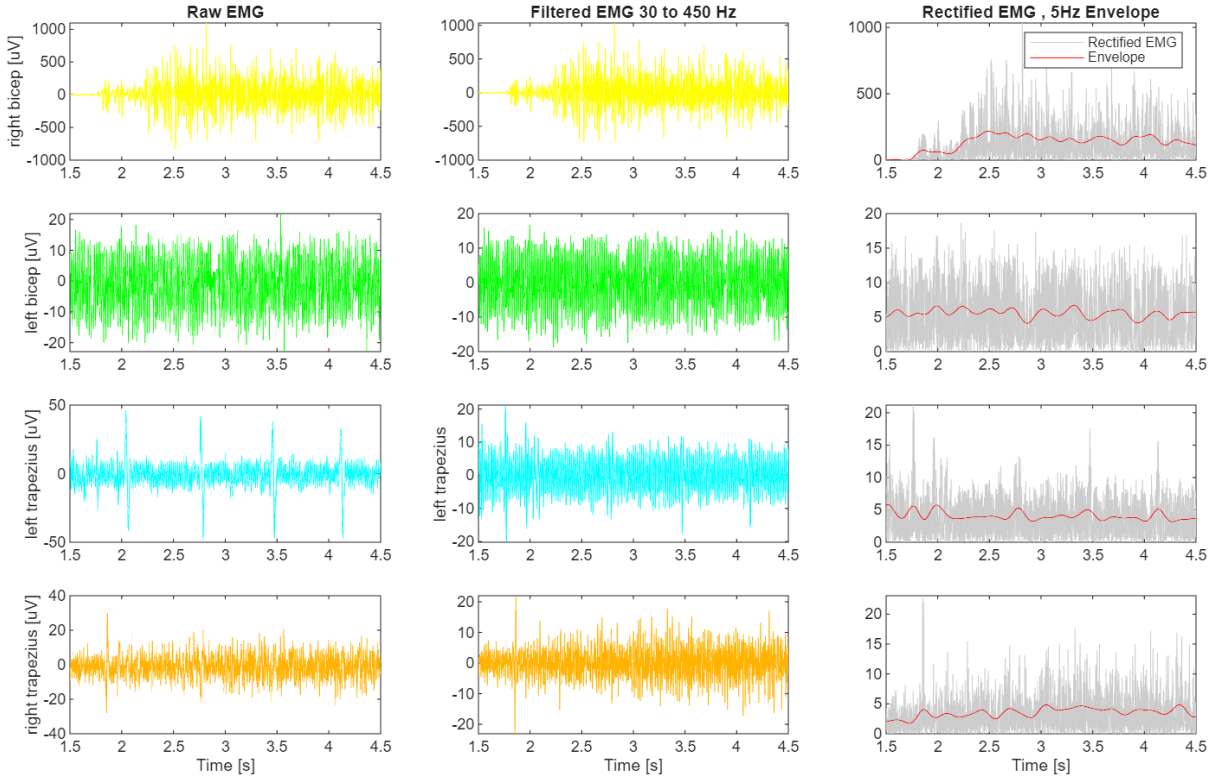


Figure 1 - Part 1 : Preprocessing Steps

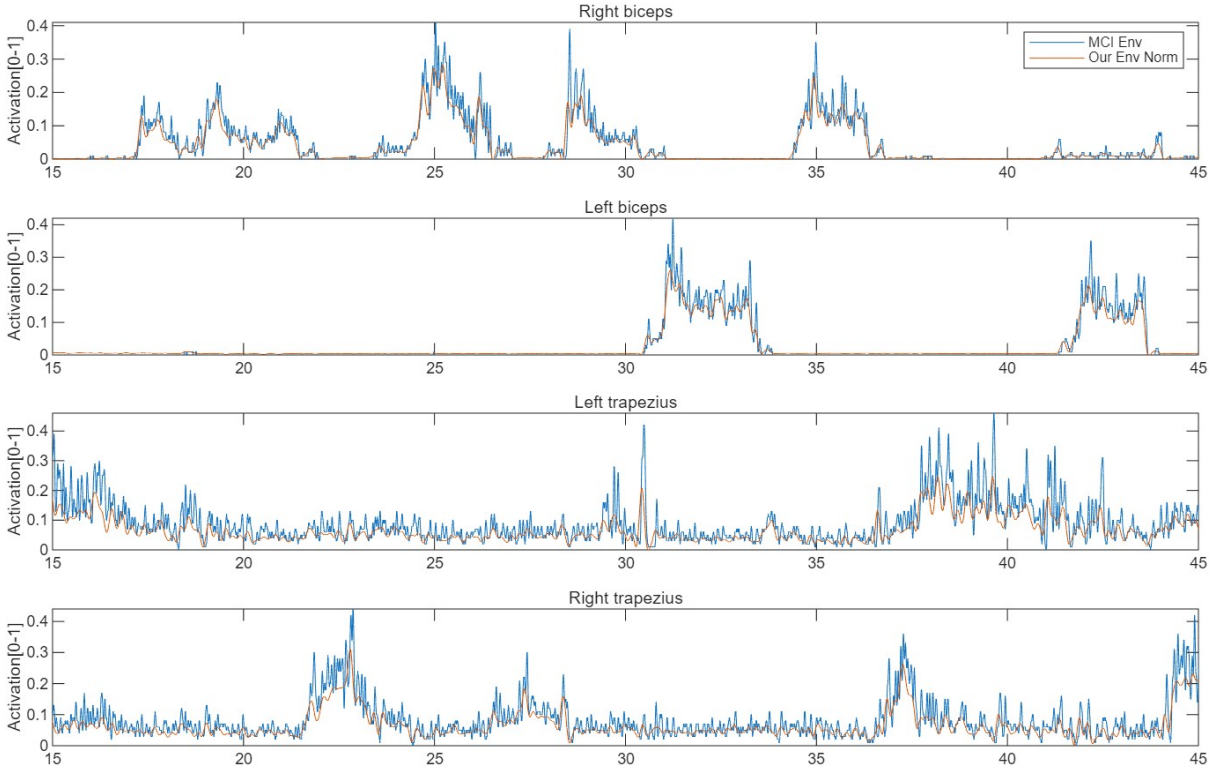


Figure 2 - Part 3 : Comparison of live recorded BoMI EMG envelopes and our Envelopes

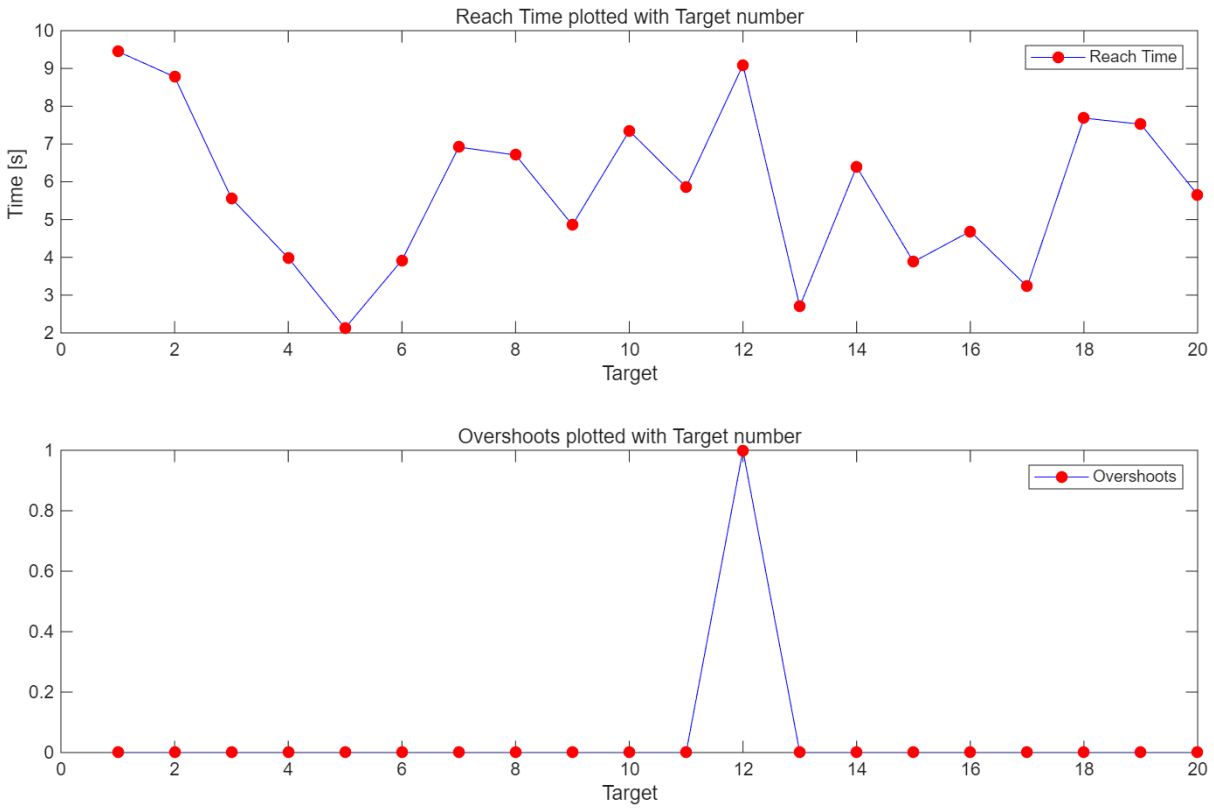


Figure 3 - Part 4 : Reach time and Overshoots plotted with Trial number

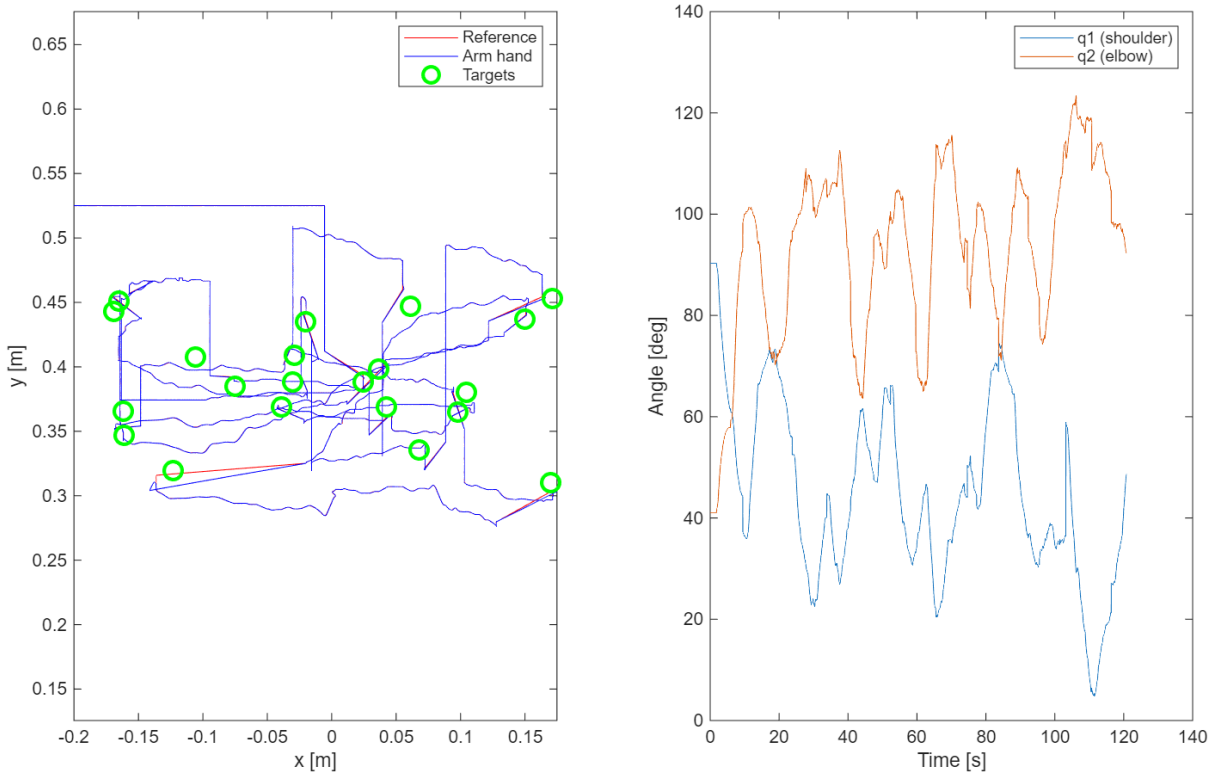


Figure 4 - Part 5 : Robot hand path and joint angles

Centerout Task results

Command window output

```
Loaded Cometa, centerout task: 518038 samples = 259.0 s, 48 targets
Muscles: R biceps, L biceps, L trapezius, R trapezius

Part 1 - Correlation and lag values
Correlation coefficient for R biceps [0 - 1] = 0.95
MCI lags our R biceps signal by 38.00 milliseconds
Correlation coefficient for L biceps [0 - 1] = 0.96
MCI lags our L biceps signal by 38.00 milliseconds

Correlation coefficient for L trapezius [0 - 1] = 0.87
MCI lags our L trapezius signal by 39.00 milliseconds

Correlation coefficient for R trapezius [0 - 1] = 0.86
MCI lags our R trapezius signal by 38.50 milliseconds

Part 2 - MCI map
x speed = 50 * (R biceps - L biceps)
y speed (down) = 50 * (R trapezius - L trapezius)

Part 3 - Cursor mean Errors
Our error mean = 0.99 cm
MCI error mean = 0.56 cm

Part 5 - Robot arm tracking
Mean tracking error = 0.08 mm
Max tracking error = 27.79 mm
Min |det(J)| = 0.0591
```

Figures

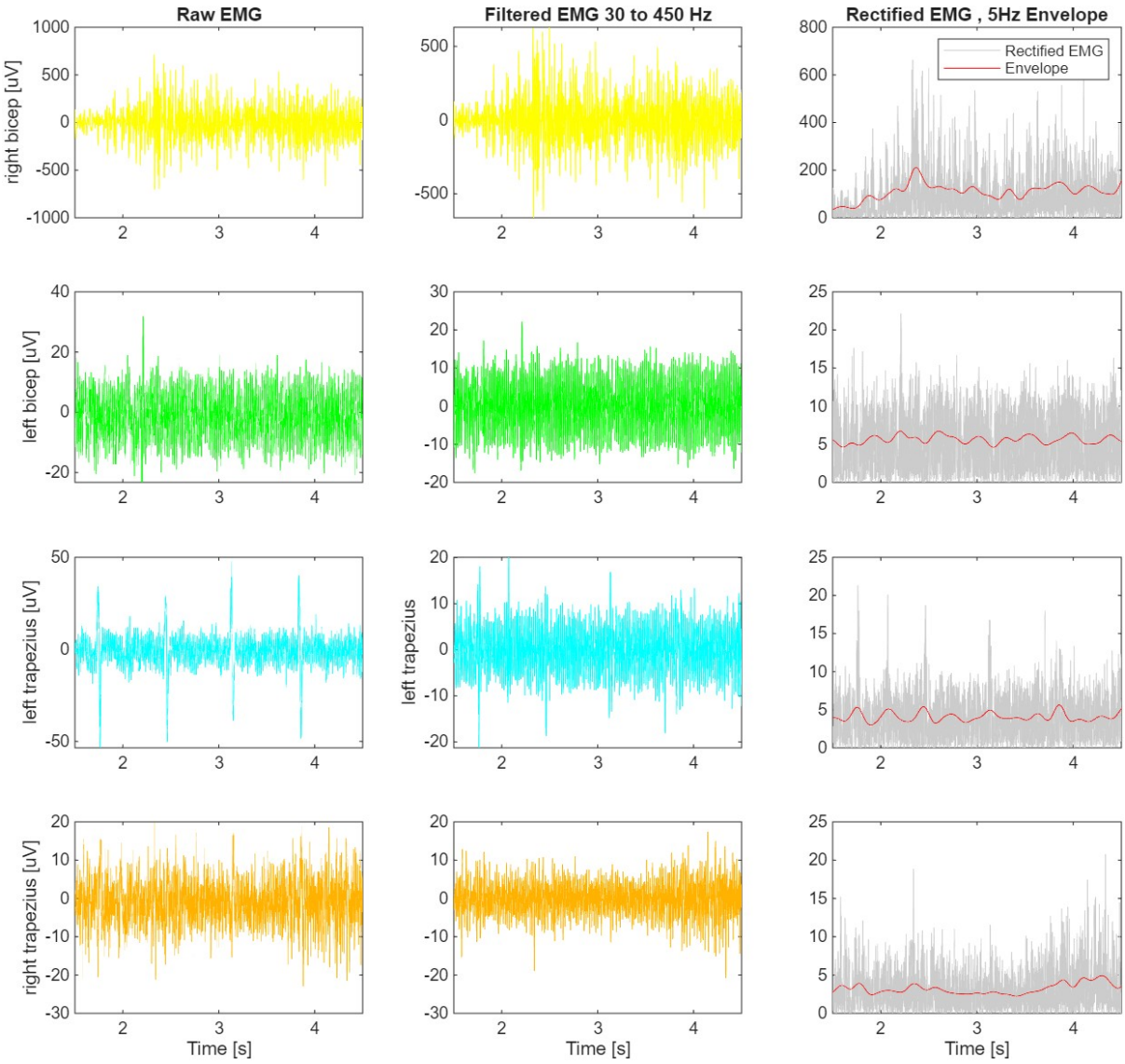


Figure 5 - Part 1 : Preprocessing Steps

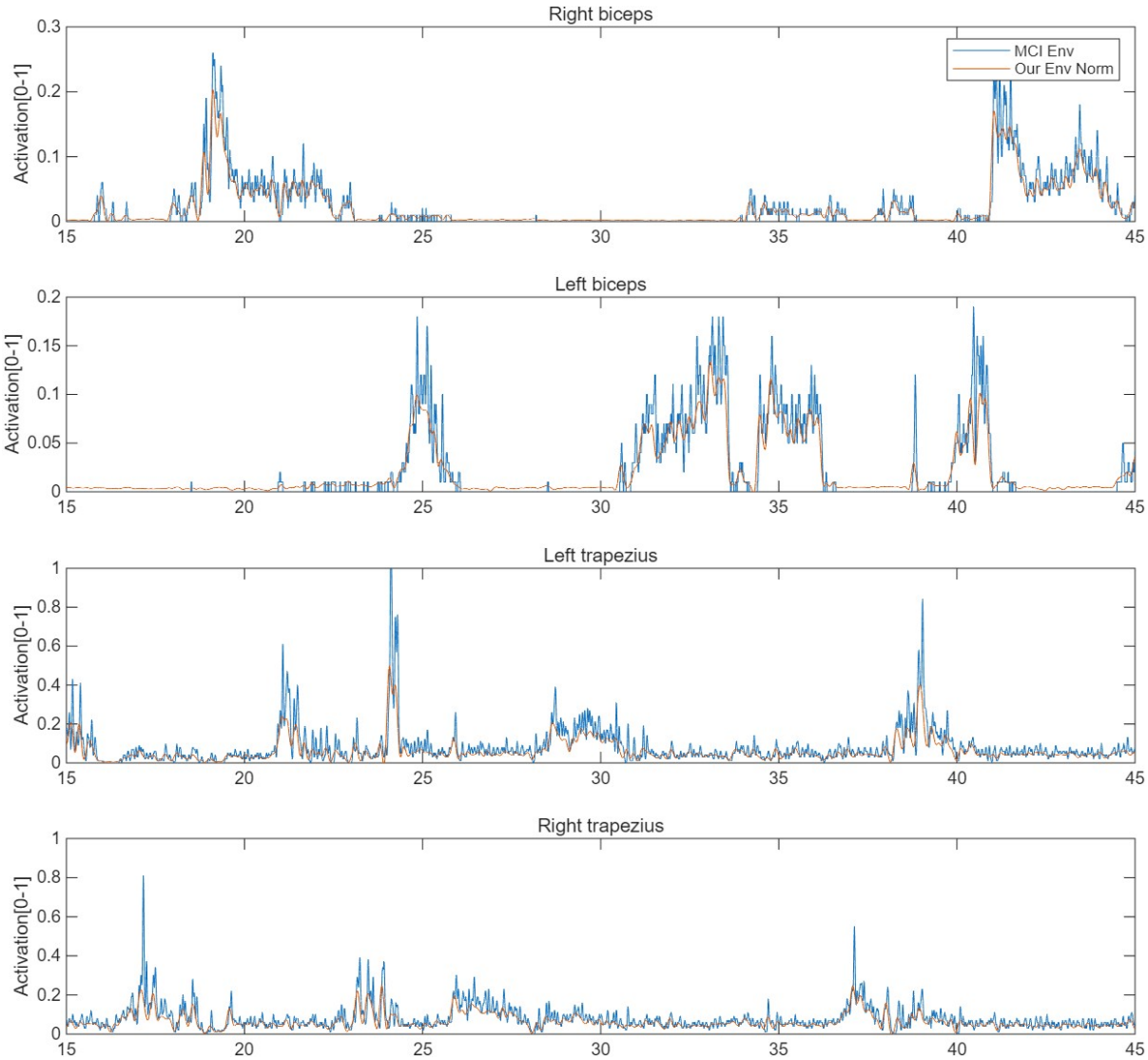
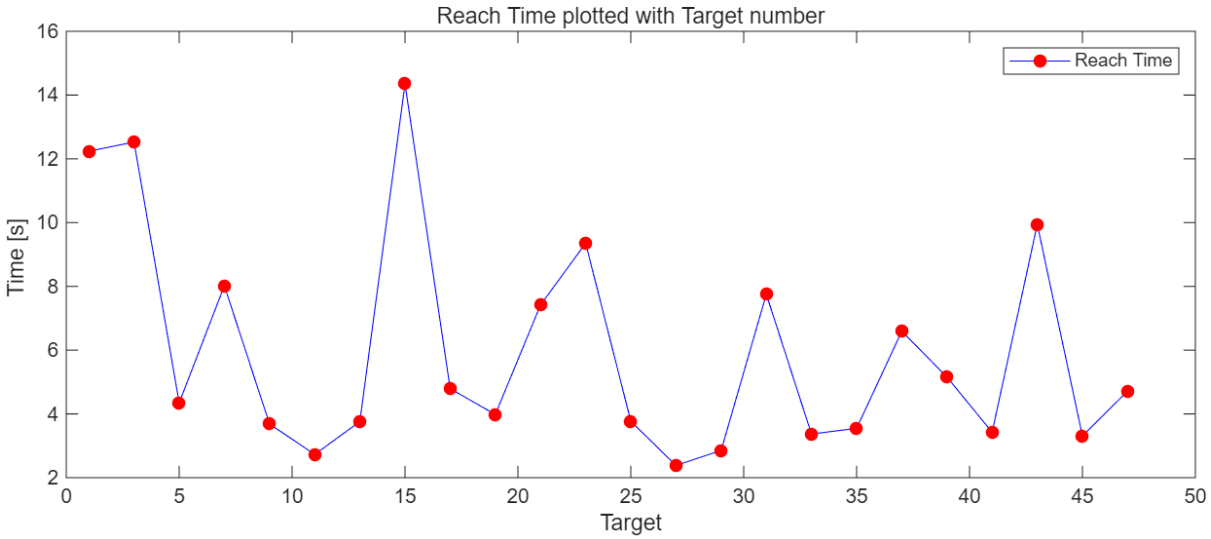


Figure 6 - Part 3 : Comparison of live recorded BoMI EMG envelopes and our Envelopes



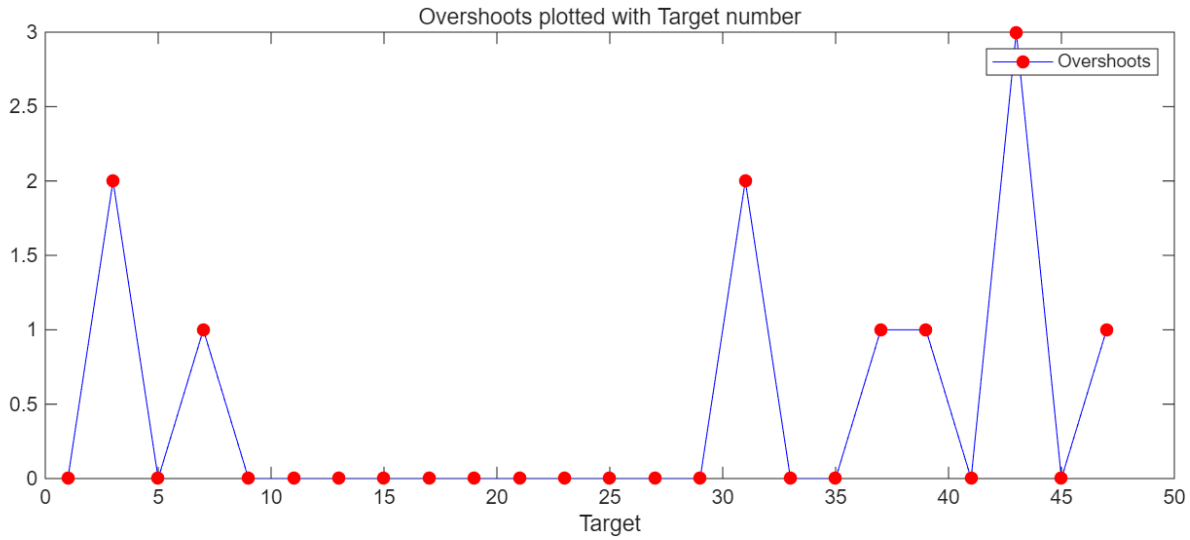


Figure 7 - Part 4 : Reach time and Overshoots plotted with Trial number

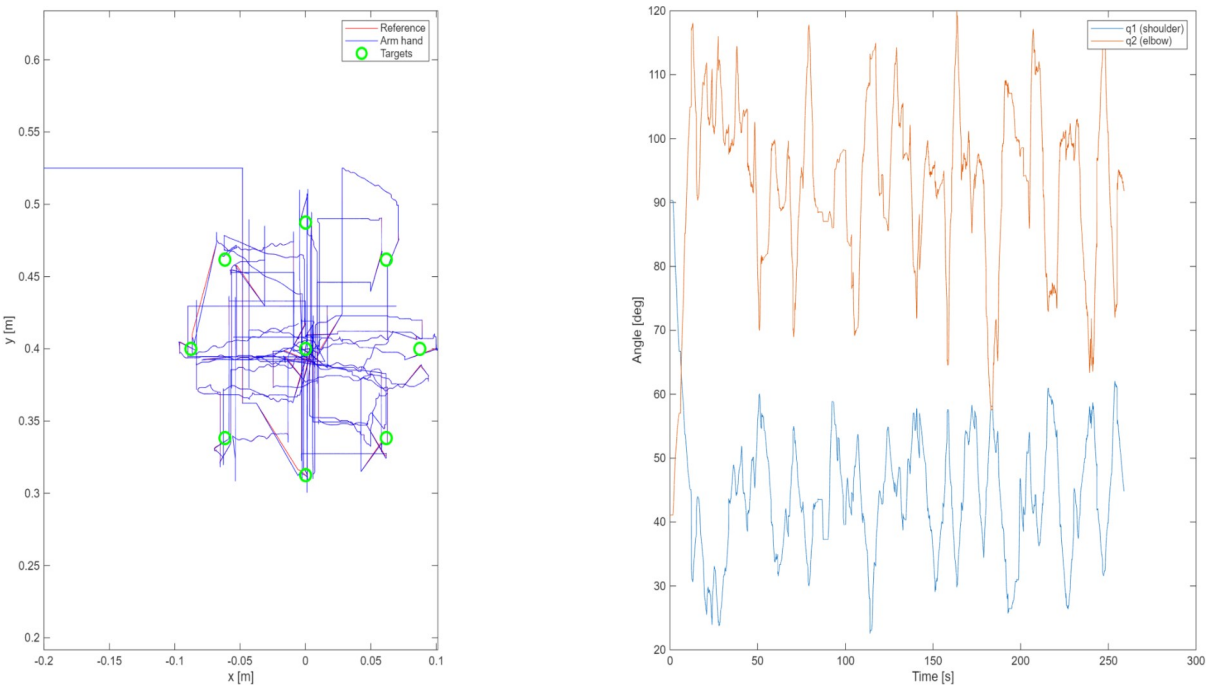


Figure 8 - Part 5 : Robot hand path and joint angles

Comments on results & figures :

Results when band pass was 30 Hz -> 450 Hz and when normalizing by envOurs/ MAX with no scaling gain. Cursor Mean Error . 2.16 cm for the center out task and 2.70 cm for the random task.

Part 1 : Correlation coefficient between bicep muscles is high accuracy , between 0.93 and 0.96 for both tasks. While the trapezius muscles have lower correlation with the live recording. It could happen due to errors in acquisition like bad placement of sEMG sensors, causing the acquisition of the signals to be worse.

The phase difference between our signal and the live recorded BoMI signal. We find out that we lead by values ranging from 36.50 ms to 41.00 ms for all muscle signals & for both tasks. Which is what is expected because the BoMI works live, hence there is a lag.

Comment on figure 1 : We saw clearly the pre processing steps : Raw signal, bandpass, Rectify then Envelope.

When experimenting with different band passes we dicsovered that using 10Hz to 450Hz instead of 30Hz to 450Hz. Improved the correlation and hence reduced the error between the live recorded BoMI and the stored EMG signal.

We multiplied the normalized EMG value by a gain which again decreased the error to 0.99 cm for the centerout task and to 0.85 cm for the random task.

And we can see that specifically the Trapezius muscles don’t have the best envelopes (not reactive enough to the bump), so maybe changing its linear envelope frequency to a higher one 6Hz to 10 Hz, could have a positive effect on the correlation and the error.

Some comments on Figure Part 3 : Comparison of live recorded BoMI EMG envelopes and the recorded Envelopes.
The lower frequency we used for the envelope is very noticable since the live BoMI used a 9Hz low pass for the evenlope, we can see that there are two major differences.

First the trade-off between a smooth envelope and a faster envelope. Second we have the small phase shift between the live signal and the recorded. The recorded leads.

1) It can be seen clearly from the figures that our signal where we reached the envelope by a lowpass filter of 5Hz was noticeably smoother than the signal recorded (envelope at 9 Hz) which make sense. As higher frequency envelopes tends to be more jittery "Jumpy" while lower frequency LP filter to get the envelope is more smooth. While the higher frequency also follows the rectified signal better. Follows the peaks and faster parts of the signal. So more reactive basically.
2) About the lag / leading of one signal w.r.to the other , there is a lead between our envelope and the recorded MCI as expected and confirmed by exact number from the correlation analysis.

Other ideas to reduce the cursor error

The purpose of this section is to test and confirm or ignore some theoritical best practicies or general ideas to further increase the correlation and reduce the error between the live recorded MCI signal and the raw recorded signal (envOurs_norm). A large majority of these ideas were not tested due to timing constraints.

- **Adding gain to the normalized envOurs_norm signal.** Specifically to the trapezius muscles the envelope doesn't seem very reactive. Additionally the correlation coefficient between the trapizius muscles is realitvely lower than the bicep muscles. Which was even noticed by Paolo who was using his muscles to control the cursor. The trapizius muscles were more difficult to move specifically the right one.

- **Introducing lag :** By using filfilt , we have zero - lag. What if we try another filter that introduces a small lag for the purpose of having both signals matching in value and phase.

By **optimizing** we mean trying many different values. While keeping the other parameters affecting the error constant and possibly in combination with other parameters. And checking the results of the error and possibly the correlation.

- **Optimizing the bandpass frequencies.** We tested this but we didn't optimize once we reached a cursor mean error of less than 1cm we were satisfied. We can further try to test different bands without going too much lower than 10Hz and not going much higher 30Hz / 450 Hz.

- **Optimizing the lowpass frequency for the envelope.** The recorded BoMI uses a 9Hz LP for the envelope while we used 5Hz. When we tried to increase it it didn't

improve the error. But we didn't optimize with trying to find the best LP frequency for the envelope.

- **Position integration at instant k instead of k + 0.5 :** In our code to calculate the position we integrate velocity at points k and sum the position (k-1) By this we have a position at instant k + 0.5 in between them! Winter suggests that to calculate the position for instant k. By using two instants K-1 and K+1 , and in this way when we integrate the two points we get the position exactly at point K.

Comments on the experiment (collection of the sEMG signals)

During the practical part, we decided that I (Paolo) would do the actual test with the sensors attached to my body. Endri and Mohamed were my "nurses", because they had to attach the sensors to the right and left biceps and the right and left trapezius. After that, we started the calibration test, where I had to contract all the muscles in order, following the computer screen.

After the first calibration, we tried a test where I had to move a cursor by contracting my muscles, but it was impossible because the cursor was moving instantly. We thought the calibration had gone badly, maybe because I was contracting the muscles too hard, so we did it again, and it failed again. At that point we thought we had placed the sensors wrong, so the teacher came to help and replaced all of them. We did the calibration again, but the test still didn't work.

Nobody understood why this was happening. After a while, the TA. noticed that the only difference from the test the TA had done in the morning was the HDMI cable connected to project the screen. Once we disconnected everything, it all worked and the game was really fun. We kept wondering why the cable was the problem, and the reason was the electrical noise it created, which interfered with the computer's program.

One last thing I noticed during the game was that my right trapezius reacted slower than the left one. This is probably the reason why the figure of MCI vs our recorded signal has that shape.